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ABSORPTION AND SCATTERING

When a beam of light is passed through matter in the solid, liquid, or gaseous state, its propagation is affected in two important ways: (1) the intensity will always decrease to a greater or lesser extent as the light penetrates farther into the medium, and (2) the velocity will be less in the medium than in free space. The loss of intensity is chiefly due to absorption, although under some circumstances scattering may play an important part. In this chapter we shall discuss the consequences of absorption and scattering, while the effect of the medium on the velocity, which comes under dispersion,

22.1 GENERAL AND SELECTIVE ABSORPTION

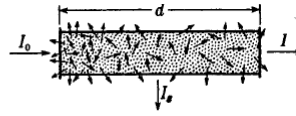
A substance is said to show *general absorption* if it reduces the intensity of all wavelengths of light by nearly the same amount. For visible light this means that the transmitted light, as seen by the eye, shows no marked color. There is merely a reduction of the total intensity of the white light, and such substances therefore appear to be gray. No substance is known which absorbs all wavelengths equally, but some, such as suspensions of lamp black or thin semitransparent films of platinum, approach this condition over a fairly wide range of wavelengths.

By *selective absorption* is meant the absorption of certain wavelengths of light in preference to others. Practically all colored substances owe their color to the existence of selective absorption in some part or parts of the visible spectrum. Thus a piece of green glass absorbs completely the red and blue ends of the spectrum, the remaining portion in the transmitted light giving a resultant sensation of green to the eye. The colors of most natural objects such as paints, flowers, etc., are due to selective absorption. These objects are said to show pigment or *body color*, as distinguished from *surface color*, since their color is produced by light which penetrates a certain distance into the substance. Then, by scattering or reflection, it is deviated and escapes from the surface, but only after it has traversed a certain thickness of the medium and has been robbed of the colors which are selectively absorbed.

22.2 DISTINCTION BETWEEN ABSORPTION AND SCATTERING

FIGURE 22A

Scattering of light by finely divided particles as in smoke.



In Fig. 22A let light of intensity I_0 enter a long glass cylinder filled with smoke. The intensity I of the beam emerging from the other end will be less than I_0 . For a given density of smoke, experiment shows that I depends on the length d of the column according to the exponential law stated in Sec. 11.9:

$$I = I_0 e^{-\alpha d} \quad (22a)$$

Here α is usually called the absorption coefficient, since it is a measure of the rate of loss of light from the direct beam. However, most of the decrease of intensity of I is in this case not due to a real disappearance of the light but results from the fact that some light is *scattered* to one side by the smoke particles and thus removed from the direct beam. Even with a very dilute smoke, a considerable intensity I_s of scattered light may easily be detected by observing the tube from the side in a darkened room. Rays of sunlight seen to cross a room from a window are made visible by the fine dust particles suspended in the air.

True *absorption* represents the actual disappearance of the light.

True *absorption* represents the actual disappearance of the light, the energy of which is converted into *heat motion* of the molecules of the absorbing material. This will occur to only a small extent in the above experiment, so that the name "absorption coefficient" for α is not appropriate in this case. In general, we can regard α as being made up of two parts, α_a due to true absorption and α_s due to scattering. Equation (22a) then becomes

$$I = I_0 e^{-(\alpha_a + \alpha_s)d} \quad (22b)$$

In many cases either α_a or α_s may be negligible with respect to the other, but it is important to realize the existence of these two different processes and the fact that in many cases both may be operating.

22.9 SCATTERING BY SMALL PARTICLES

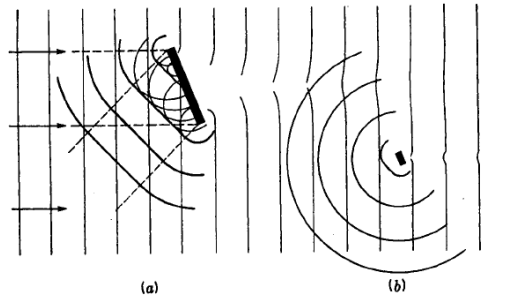


FIGURE 22F
The reflection and diffraction of light by small objects comparable in size with the wavelength of light.

In (b) of the figure, the reflector is much smaller than a wavelength, and here the spreading is so great that the reflected waves differ very little from uniform spherical waves. In this case the light taken from the primary beam is said to be scattered, rather than reflected, since the law of reflection has ceased to be applicable. Scattering is therefore a special case of diffraction. The wave scattered from an object much smaller than a wavelength of light will be spherical, regardless of whether or not the object has the plane form assumed in Fig. 22F(b). This follows from the fact that there can be no interference between the wavelets emitted by the several points on the surface of the scattering particle, inasmuch as the extreme points are separated by a distance much less than the wavelength.

The first quantitative study of the laws of scattering by small particles was made in 1871 by Rayleigh,* and such scattering is frequently called *Rayleigh scattering*.

The mathematical investigation of the problem gave a general law for the intensity of the scattered light, applicable to any particles of index of refraction different from that of the surrounding medium. The only restriction is that the linear dimensions of the particles be considerably smaller than the wavelength. As we might expect, the scattered intensity is found to be proportional to the incident intensity and to the square of the volume of the scattering particle. The most interesting result, however, is the dependence of scattering on wavelength. With a given size of the particles, long waves would be expected to be less effectively scattered than short ones, because the particles present obstructions to the waves which are smaller *compared with the wavelength* for long waves than for short ones. In fact, as will be proved in Sec. 22.13, the intensity is proportional to $1/\lambda^4$:

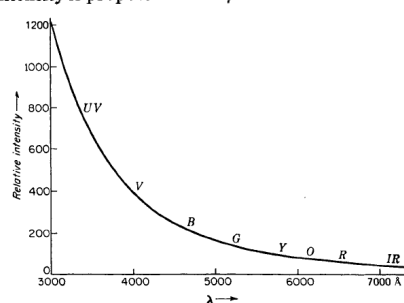


FIGURE 22G
Intensity of scattering versus wavelength according to Rayleigh's law.

$$I_s = k \frac{1}{\lambda^4}$$

22.13 SCATTERING AND REFRACTIVE INDEX

The fact that the velocity of light in matter differs from that in vacuum is a consequence of scattering. The individual molecules scatter a certain part of the light falling on them, and the resulting scattered waves *interfere* with the primary wave, bringing about a change of phase which is equivalent to an alteration of the wave velocity.

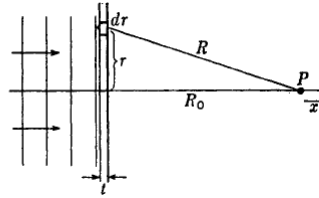


FIGURE 22H
Geometry of scattering by a thin lamina.

In Fig. 22H plane waves are shown striking an infinitely wide sheet of a transparent material, the thickness of which is small compared to the wavelength. Let the electric vector in this incident wave have unit amplitude, so that in the exponential notation (Sec. 14.8) it may be represented at a particular time by $E = e^{ikx}$. If the fraction of the wave that is scattered is small, the disturbance reaching some point P will be essentially the original wave, plus a small contribution due to the light scattered by all the atoms in the thin lamina. To evaluate the latter, we note that its intensity is proportional to the coefficient α_s of Eq. (22b). This measures the fractional decrease of intensity by scattering in traversing the small thickness t , to which the scattered intensity must be proportional. We therefore have

$$-\frac{dI}{I} = \alpha_s t \approx I_s \quad (22c)$$

The intensity scattered by a single atom, since there are Nt atoms per unit area of the lamina, becomes

$$I_1 \approx \frac{\alpha_s t}{Nt} = \frac{\alpha_s}{N}$$

and the amplitude

$$E_1 \approx \sqrt{\frac{\alpha_s}{N}}$$

the total scattered amplitude

$$E_s \approx Nt \sqrt{\frac{\alpha_s}{N}} = t \sqrt{\alpha_s N}$$

The complex amplitude at P is obtained by integrating this quantity over the surface of the lamina, and adding it to the amplitude of the primary wave. The resultant then becomes

$$E + E_s = e^{ikR_0} + t\sqrt{\alpha_s N} \int_0^\infty \frac{2\pi r}{R} e^{ikR} dr$$

where the factor $1/R$ enters because of the inverse-square law. Now since $R^2 = R_0^2 + r^2$, we have $r dr = R dR$, and the integral may be written

$$\int_0^\infty \frac{2\pi}{R} e^{ikR} r dr = 2\pi \int_{R_0}^\infty e^{ikR} dR = \frac{2\pi}{ik} [e^{ikR}]_{R_0}^\infty$$

Since the wave trains always have a finite length, the scattering as $R \rightarrow \infty$ can contribute nothing to the coherent wave. Substituting the lower limit of the integral, we find

$$\begin{aligned} E + E_s &= e^{ikR_0} - t\sqrt{\alpha_s N} \frac{\lambda}{i} e^{ikR_0} \\ &= e^{ikR_0} + t\sqrt{\alpha_s N} i \lambda e^{ikR_0} \\ &= e^{ikR_0} (1 + i\lambda t\sqrt{\alpha_s N}) \end{aligned}$$

By our original assumption, the second term in parentheses is small compared with the first. These will be recognized as the first two terms in the expansion of $e^{i\lambda t\sqrt{\alpha_s N}}$, and may here be equated to it, giving

$$E + E_s = \exp ikR_0 \exp (i\lambda t\sqrt{\alpha_s N}) = \exp [i(kR_0 + \lambda t\sqrt{\alpha_s N})]$$

Thus the phase of the wave at P has been altered by the amount $\lambda t\sqrt{\alpha_s N}$. But we know (Sec. 13.15) that the presence of a lamina of thickness t and refractive index n gives a phase retardation of $(2\pi/\lambda)(n - 1)t$. Hence

$$\lambda t\sqrt{\alpha_s N} = \frac{2\pi}{\lambda} (n - 1)t$$

and finally

$$n - 1 = \frac{\lambda^2}{2\pi} \sqrt{\alpha_s N} \quad (22d)$$

This important relation contains Rayleigh's law of scattering

by Eq. (22c), I_s is proportional to α_s , this scattered intensity varies as $1/\lambda^4$, assuming n to be independent of wavelength. In our derivation no absorption has been considered, so that the equation is valid only for wavelengths well away from any absorption bands. In the next chapter we shall see how the refractive index behaves as the wavelength approaches that of an absorption band.

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THE POLARIZATION OF LIGHT

From the properties of interference and diffraction we are led to conclude that light is a wave phenomenon, and we utilize these properties to measure the wavelength. These effects tell us nothing about the types of waves with which we are dealing—whether they are longitudinal or transverse, or whether the vibrations are linear, circular, or torsional. The electromagnetic theory, however, specifically requires that the vibrations be transverse, being therefore entirely confined to the plane of the wave front. The most general type of vibration is elliptical, of which linear and circular vibrations are extreme cases. Experiments which bring out these characteristics are those dealing with the polarization of light. Although a longitudinal wave like a sound wave must necessarily be symmetrical about the direction of its propagation, transverse waves may show dissymmetries, and if any beam of light shows such a dissymmetry, we say it is polarized.

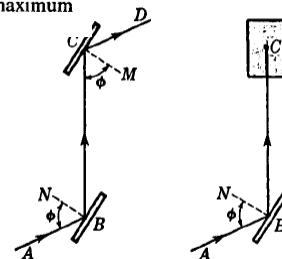
the various polarizing devices act to separate ordinary light into its polarized components. The common methods used in producing and demonstrating the polarization of light may be grouped under the following heads: (1) reflection, (2) transmission through a pile of plates, (3) dichroism, (4) double refraction, and (5) scattering.

24.1 POLARIZATION BY REFLECTION

A beam of unpolarized light, AB in Fig. 24A, is incident at an angle of about 57° on the first glass surface at B . This light is again reflected at 57° by a second glass plate C placed parallel to the first as shown at the left. If now the upper plate is rotated about BC as an axis, the intensity of the reflected beam is found to decrease, reaching zero for a rotation of 90° . Rotation about BC keeps the angle of incidence constant. The experiment is best performed with the back surfaces of the glass painted black. The first reflected beam BC' then appears to be cut off and to vanish at C' . As the upper mirror is rotated further about BC the reflected beam CD reappears, increasing in intensity to reach a maximum at 180° . Continued rotation produces zero intensity again at 270° , and a maximum again at 360° , the starting point.

the angle of incidence ϕ *polarizing angle*

FIGURE 24A
Polarization by reflection from glass surfaces.



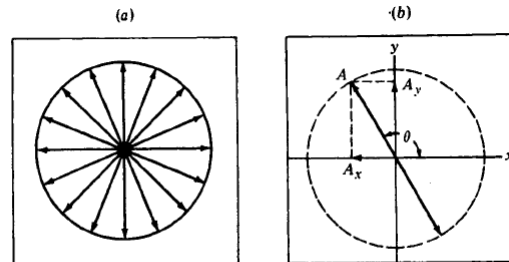


FIGURE 24B
Vibrations in unpolarized light viewed end-on. (a) All planes are equally probable. (b) Each vibration can be resolved into two components in the x and y directions.

24.3 POLARIZING ANGLE AND BREWSTER'S LAW

Consider unpolarized light to be incident at an angle ϕ on a dielectric like glass, as shown in Fig. 24D(a). There will always be a reflected ray OR and a refracted ray OT . An experiment like the one described in Sec. 24.1 and shown in Fig. 24A shows that the reflected ray OR is partially plane-polarized and that only at a certain definite angle, about 57° for ordinary glass, is it plane-polarized. It was Brewster who first discovered that at this polarizing angle $\bar{\phi}$ the reflected and refracted rays are just 90° apart. This remarkable discovery enables one to correlate polarization with the refractive index

$$\frac{\sin \phi}{\sin \phi'} = n \quad (24a)$$

Since at $\bar{\phi}$ the angle $ROT = 90^\circ$, we have $\sin \bar{\phi}' = \cos \bar{\phi}$, giving

$$\frac{\sin \bar{\phi}}{\sin \bar{\phi}'} = \frac{\sin \bar{\phi}}{\cos \bar{\phi}} = n$$

$$\bullet \quad n = \tan \bar{\phi} \quad (24b)$$

This is *Brewster's law*, which shows that the angle of incidence for maximum polarization depends only on the refractive index. It therefore varies somewhat with wavelength, but for ordinary glass the dispersion is such that the polarizing angle $\bar{\phi}$ does not change much over the whole visible spectrum. This fact is readily verified by calculating $\bar{\phi}$ for several wavelengths, using the values of n from Table 23B,

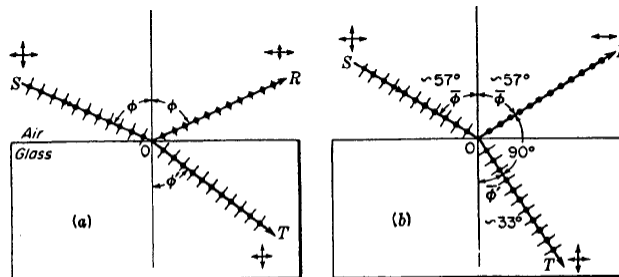


FIGURE 24D
(a) Polarization by reflection and refraction. (b) Brewster's law for the polarizing angle.

24.4 POLARIZATION BY A PILE OF PLATES

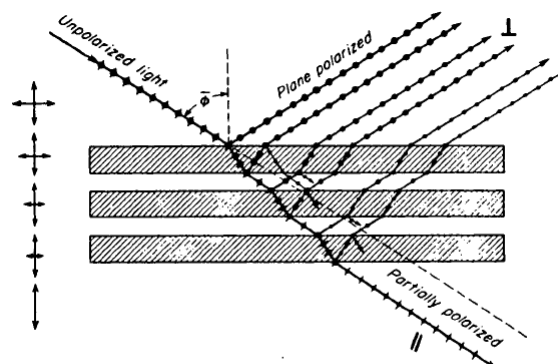


FIGURE 24E
Polarization of light by a pile of glass plates.

For a single surface of glass with $n = 1.50$, at the polarizing angle 100 percent of the light vibrating parallel to the plane of incidence is transmitted, whereas for the perpendicular vibrations only 85 percent is transmitted, the other 15 percent being reflected. Obviously the degree of polarization of the transmitted beam is small for a single surface.

The degree of polarization P of the transmitted light can be calculated by summing the intensities of the parallel and perpendicular components. If these intensities are called I_p and I_s , respectively, it has been shown* that

$$P = \frac{I_p - I_s}{I_p + I_s} = \frac{m}{m + [2n^2/(1 - n^2)]} \quad (24c)$$

where m is the number of plates, that is, $2m$ surfaces, and n their refractive index.

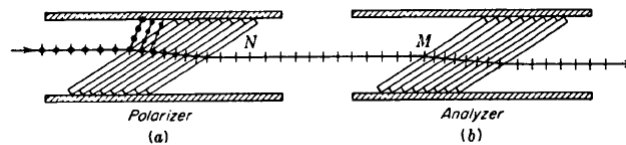


FIGURE 24F
Glass plates mounted at the polarizing angle $\bar{\phi}$.

The pile of plates may be used, however, to illustrate a common arrangement in the production and analysis of polarized light.

24.5 LAW OF MALUS*

This law tells us how the intensity transmitted by the analyzer varies with the angle that its plane of transmission makes with that of the polarizer. In the case of two piles of plates, the plane of transmission is the plane of incidence, and for the law of Malus to hold we must assume that the transmitted light is completely plane-polarized.

Then the law of Malus states that the transmitted intensity varies as the square of the cosine of the angle between the two planes of transmission.

The amplitude of the light that passes through the analyzer is $A_1 = A \cos \theta$

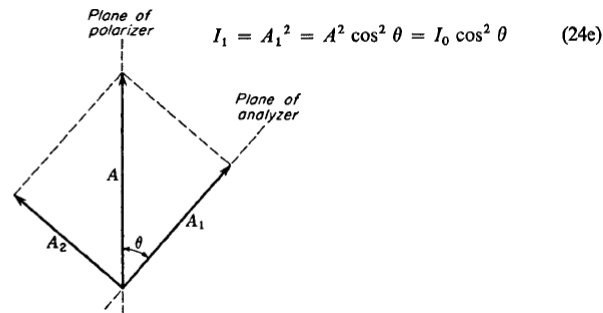


FIGURE 24G
Resolution into components of the amplitude of plane-polarized light.

24.6 POLARIZATION BY DICHROIC CRYSTALS

These crystals have the property of selectively absorbing one of the two rectangular components of ordinary light. Dichroism is exhibited by a number of minerals and by some organic compounds. Perhaps the best known of the mineral crystals is *tourmaline*.

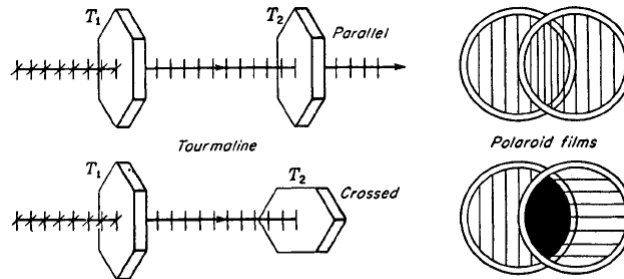


FIGURE 24H
Dichroic crystals and polarizing films in the *parallel* and *crossed* positions.

24.7 DOUBLE REFRACTION

24.10 POLARIZATION BY DOUBLE REFRACTION

FIGURE 24I
Calcite and quartz crystal forms. The direction of the optic axis is indicated by broken lines.

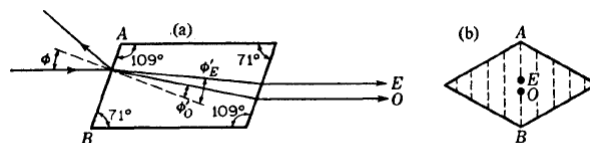
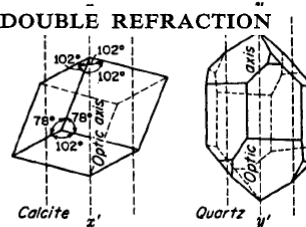


FIGURE 24J
Side and end views of the double refraction of light by a calcite crystal. (a) Cross-section of a principal plane. (b) End view.

angles of refraction ϕ' for different angles of incidence ϕ , one finds that Snell's law of refraction

$$\frac{\sin \phi}{\sin \phi'} = n \quad (24f)$$

holds for one ray but not for the other. The ray for which the law holds is called the *ordinary* or *O* ray, and the other is called the *extraordinary* or *E* ray.

NICOL PRISM
CALCITE PRISMS
ROCHON AND WOLLASTON PRISMS

24.11 NICOL PRISM

This very useful polarizing device is made from a calcite crystal, and derives its name from its inventor.* The nicol prism is made in such a way that it removes one of the two refracted rays by total reflection, as is illustrated in Fig. 24M. There are several different forms of nicol prism,† but we shall describe here one of the commonest ones. First a crystal about 3 times as long as it is wide is taken and the ends cut down from 71° in the principal section to a more acute angle of 68° . The crystal is then cut apart along the plane $A'D'$ perpendicular to both the principal section and the end faces. The two cut surfaces are ground and polished optically flat and then cemented together with canada balsam. Canada balsam is used because it is a clear transparent substance with an index of refraction about midway between the index of the O and E rays. For sodium light,

Index of O ray	$n_o = 1.65836$
Index of canada balsam	$n_B = 1.55$
Index of E ray	$n_E = 1.48641$

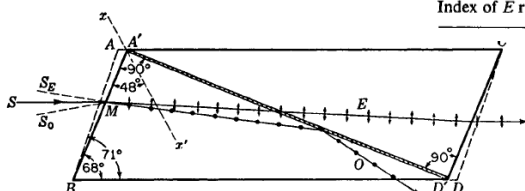


FIGURE 24M
Detailed diagram of a nicol prism, showing how it is made from a calcite crystal.

Optically the balsam is denser than the calcite for the E ray and less dense for the O ray. The E ray therefore will be refracted into the balsam and on through the calcite crystal, whereas the O ray for large angles of incidence will be totally reflected. The critical angle for total reflection of the O ray at the first calcite to balsam surface is about 69° and corresponds to a limiting angle SMS_o in Fig. 24M of about 14° . At greater angles than this, some of the O ray will be transmitted. This means that a nicol should not be used in light which is highly convergent or divergent.

The E ray in a nicol also has an angular limit, beyond which it will be totally reflected by the balsam. This is due to the fact that the index of refraction of calcite is different for different directions through the crystal. In the next chapter it will be seen that the index $n_E = 1.486$, as it is usually given, applies only in the special case of light traveling at right angles to the optic axis. Along the optic axis the E ray travels with the same speed as the O ray, and it therefore has the same index of 1.658. For intermediate angles the effective index lies between these two limits 1.486 and 1.658. There will therefore be a maximum angle SMS_E beyond which the balsam will be optically less dense than the calcite, and there will be total reflection of the E vibrations. The prism is so cut that this angle likewise is in the neighborhood of 14° . The direction of the incident light on a nicol therefore is limited on the one side to avoid having the O ray transmitted and on the other side to avoid having the E ray totally reflected. In practice, it is important to keep this limitation in mind.